MUIZ ABIOLA AMINU

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RESEARCH INTERESTS

Control Systems, Embedded Architectures, Autonomous Mobile Robotics, Edge Intelligence (TinyML), and Mechatronic Integration for Industrial, Renewable Energy Systems.

EDUCATION

Federal University of Technology, Minna — Minna, Nigeria
Bachelor of Engineering (B.Eng.) in Mechatronics Engineering — Dec 2025
Second Class Honours (Upper Division) | **CGPA:** 4.37 / 5.00 (87%)
Capstone Thesis: *Development of a Semi-Autonomous Navigation System for a Pipeline Monitoring Robot*

PUBLICATIONS & CONFERENCE PRESENTATIONS

- **Aminu, M. A.** (2026). "Development of a Semi-Autonomous Navigation System for a Pipeline Monitoring Robot." *Preprint.* [**Preprint PDF**]
- **Aminu, M. A.** (2026). "Hardware-in-the-Loop Benchmarking of Edge-NILM on RISC-V Silicon: Decentralized Demand-Side Management for Resilient African Solar Mini-Grids." *Accepted for Poster Presentation, BUILD26 Conference*, Abuja, Nigeria. [**Paper PDF**]

RESEARCH & ENGINEERING EXPERIENCE

EN1 Technologies | *Embedded Systems Engineer* — Jun 2026 – Present
- Architect bare-metal STM32F407 firmware for e-tricycle ECUs and develop ESP32 thermoforming control systems integrating SSR loops and MQTT telemetry.

Aflon Center for Artificial Intelligence and Robotics | *IoT Research Engineer* — Jan 2026 – Jun 2026
- Prototyped IoT hardware and designed architectures bridging microcontroller telemetry with cloud platforms.

Advanced Engineering Innovation Research Group | *Research Assistant* — Jul 2021 – Dec 2025
- Evaluated experimental setups for autonomous obstacle detection, sensor fusion, and wireless telemetry links.

Ideal Intelligent Innovation Research Group | *Research Associate* — May 2022 – Dec 2025
- Prototyped hardware and optimized motor control logic for automation projects while mentoring student researchers.

Artificial Intelligence for Clean Energy Ltd. | *Engineering Intern* — Jun 2024 – Nov 2024
- Engineered smart energy telemetry hardware for clean energy monitoring and microgrid applications.

E-lord Technology | *Hardware Engineer (Part-Time)* — Aug 2019 – Present
- Execute schematic capture, footprint assignment, and multi-layer PCB layout utilizing EasyEDA for prototypes.

SELECTED ENGINEERING PROJECTS

- **Semi-Autonomous Pipeline Inspection Robot:** Engineered an inspection rover using Raspberry Pi 4, featuring ultrasonic hazard detection (1.02 s halt response) and bidirectional RESTful API teleoperation via React/Node.js.
- **Edge AI Microcontroller Benchmarking:** Benchmarked TensorFlow Lite for Microcontrollers (TinyML) inference latency, memory footprint, and int8 quantization performance across ESP32-C3 architectures.
- **Life-Guard Node (Symbiotic Pipeline Security):** Engineered a solar-powered telemedical proof-of-concept integrating Gemini AI for predictive health diagnostics of the pipeline host communities. Designed a LoRa safety interlock that autonomously severs power and dispatches alerts during simulated pipeline breaches.

TECHNICAL SKILLS

- **Embedded Systems & SBCs:** ARM Cortex-M (STM32F407), ESP32, AVR, Raspberry Pi.
- **Firmware & Programming:** Bare-metal C/C++, CMSIS, FreeRTOS, Python, Dart (Flutter), MATLAB/Simulink.
- **Hardware Design:** Multi-layer PCB Layout (KiCad, EasyEDA), Stepper/Servo Drivers, SSRs, Power Electronics.
- **IoT Protocols:** MQTT (Mosquitto), REST APIs, Firebase, Node-RED, UART, SPI, I2C.

CERTIFICATIONS & AFFILIATIONS

- **Memberships:** Professional Member, Society of Petroleum Engineers (SPE); Graduate Member, Nigerian Society of Engineers (GMNSE).
- **Certifications:** Simulink Fundamentals (MathWorks); Career Essentials in Sustainable Tech (Microsoft).
- **Leadership:** Lead STEM Facilitator (Build-a-thon, 2024–2025) – instructed 250+ students in embedded hardware.